Problem 1 - The Talking Toy Box

Week 7 | Category B Practice | Java solutions

Problem

Toy is abstract, its toyId is final and generated from a shared static counter. ToyCar and ToyRobot directly extend Toy.

Complete Java Code

```
abstract class Toy {
    private static int nextId = 1001;
    private final String toyId;

    protected Toy() {
        toyId = "TOY-" + nextId++;
    }

    public abstract String makeSound();

    public String getToyId() {
        return toyId;
    }
}

class ToyCar extends Toy {
    private final String name;
    public ToyCar(String name) { super(); this.name = name; }
    @Override public String makeSound() { return name + ": Vroom vroom!"; }
}

class ToyRobot extends Toy {
    private final String name;
    public ToyRobot(String name) { super(); this.name = name; }
    @Override public String makeSound() { return name + ": Beep boop!"; }
}

public class Main {
    public static void main(String[] args) {
        ToyCar c = new ToyCar("Speedster");
        ToyRobot r = new ToyRobot("Bolt");
        System.out.println(c.makeSound());
        System.out.println(r.makeSound());
        System.out.println(c.getToyId());
        System.out.println(r.getToyId());
    }
}
```

Sample Output

```
Speedster: Vroom vroom!
Bolt: Beep boop!
TOY-1001
TOY-1002
```

How it works

Toy is abstract, so new Toy() is illegal. Both subclasses call the Toy constructor via super(), where the static counter is incremented exactly once per object.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.